APPARATUS RESPIRATORIUS - RESPIRATORY SYSTEM

Breathing (respiration)
Gas exchange

- O_2 & CO_2
- Gases pass within the respiratory tract

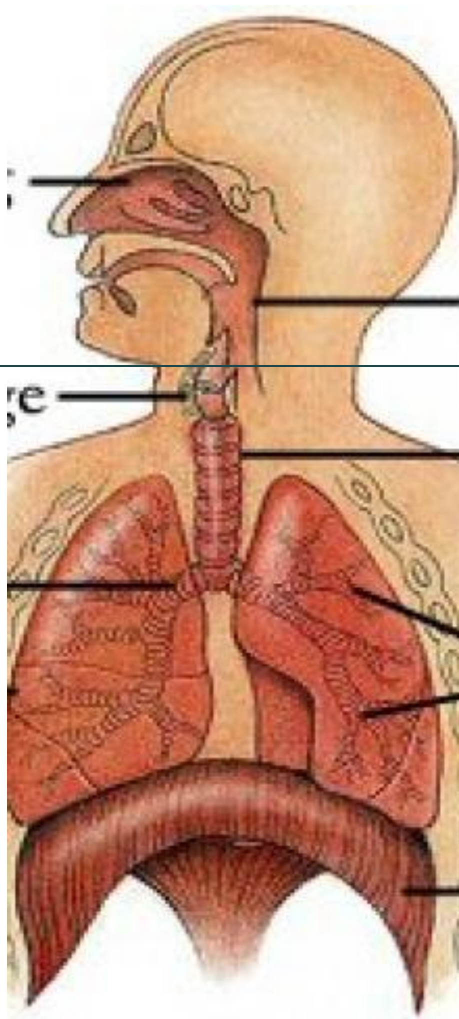
- ***External respiration:*** gas exchange between air and blood
- ***Internal respiration:*** between blood and body cells

Vocalisation (phonation)

Sense of smell (olfaction)

APPARATUS RESPIRATORIUS - RESPIRATORY SYSTEM

Conducting part vs *respiratory part*



Upper airways

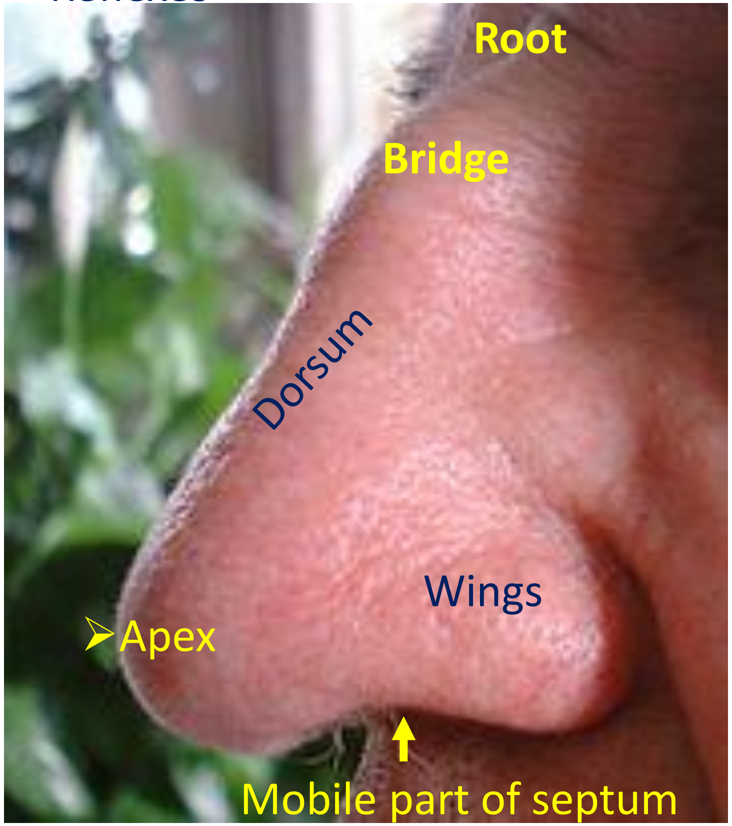
Nasus externus
Cavum nasi
Sinus paranasales
Pars nasalis pharyngis
Pars oralis pharyngis
Pars laryngea pharyngis

Lower airways

Larynx
Trachea
Bronchi principales
Pulmo dexter et sinister
„Bronchial tree” (bronchi lobares, bronchi segmentales, bronchi, bronchioli, ductus alveolares)

THE NOSE („*nasus externus*”)

- Warming/cooling & moistening of inspired air
- Mucociliary transport
- Mucosal barrier (defence mechanism)
- Resonance
- Olfaction
- Reflexes



COMPONENTS

- Cartilage
  - Lateral nasal
  - Greater and lesser alar
  - Nasal septal
  - Accessory
- Bone
  - Nasal (+maxilla+frontal)
- Muscles
  - nasalis
  - levator labii sup. alequae nasi
  - procerus
- Dense connective tissue (CT)
- Skin and /or tunica musosa

Nares /nostrils

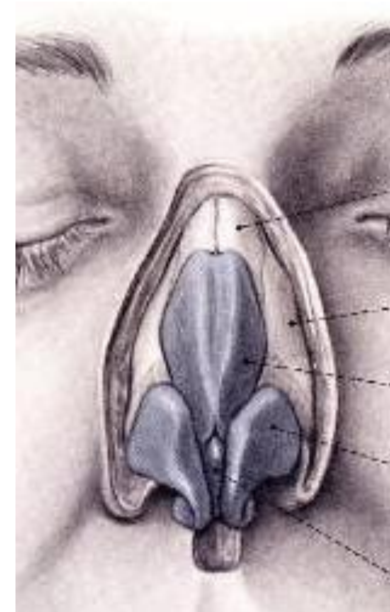
Vestibule – **limen nasi**

Cavum/cavity

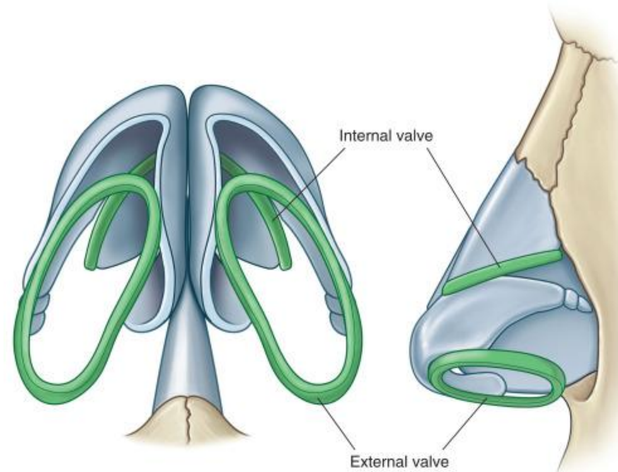
Choanae

THE EXTERNAL NOSE

- Osseous pyramid composed of *frontal* and *nasal* bones + the *maxillae*.
- Cartilagineous „dorsum”, formed by the *septal* and *dorsal cartilages*
- The tip (apex) is composed of the *alar* and *septal cartilages + CT*
- *The diameter of the nostrils define the size of the vestibule*

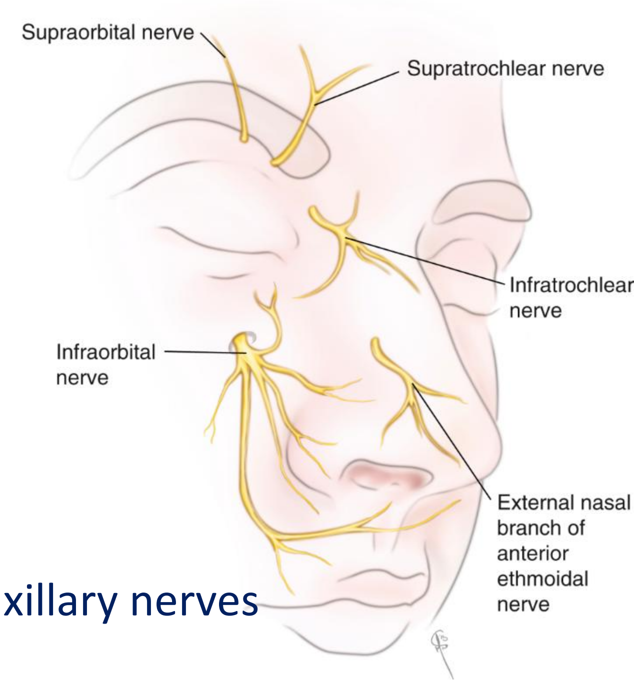


External and internal valves



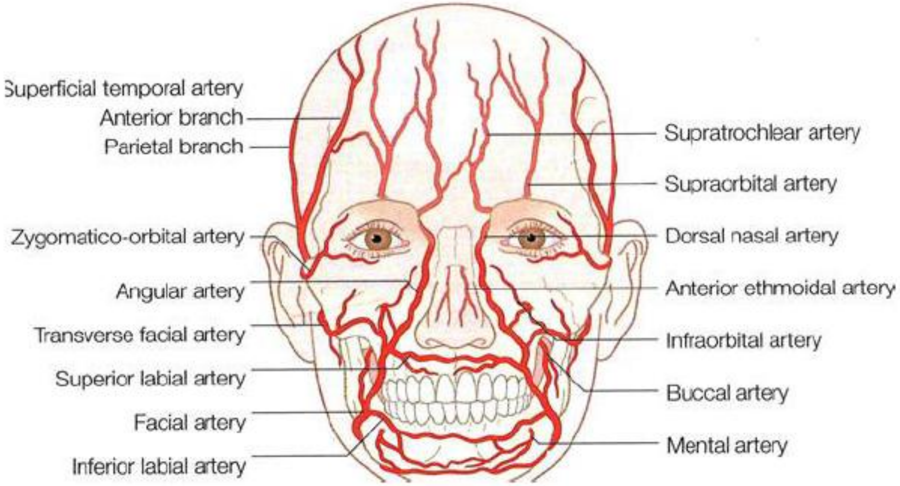
CUTANEOUS INNERVATION

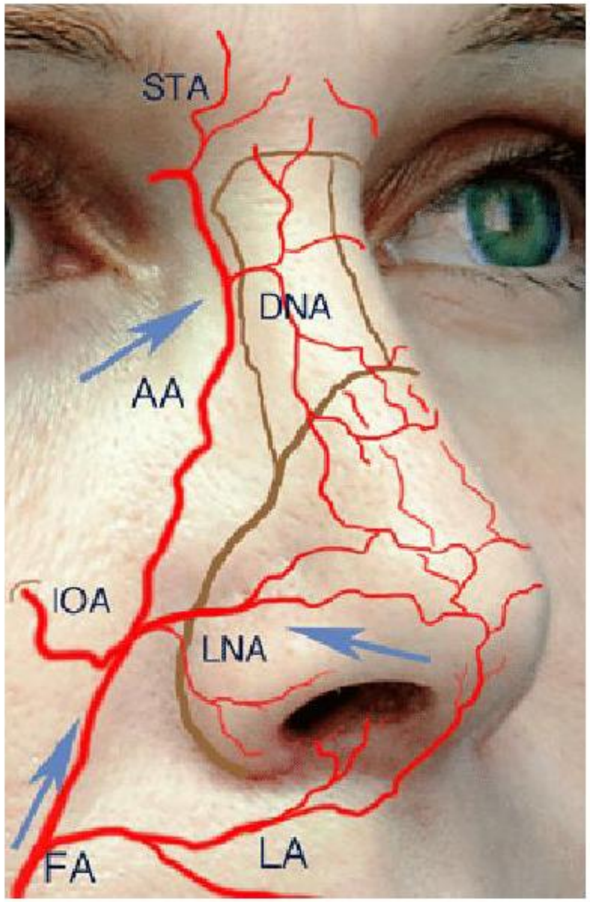
➢Branches of ophthalmic & maxillary nerves



ARTERIOUS SUPPLY

given by **the facial artery, becoming the angular artery. The alar and dorsal regions of the nose are supplied by the infraobrital branches of the maxillary artery (*external carotid*) and ophthalmic arteries (*internal carotid* system)**





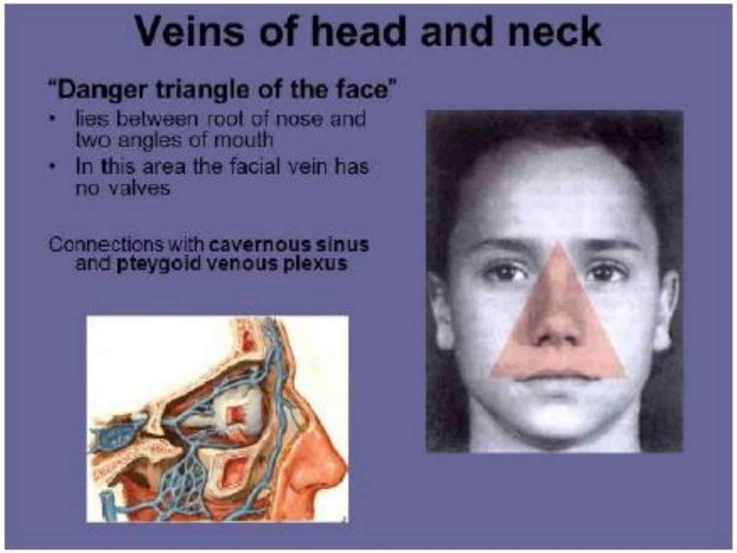
ECA

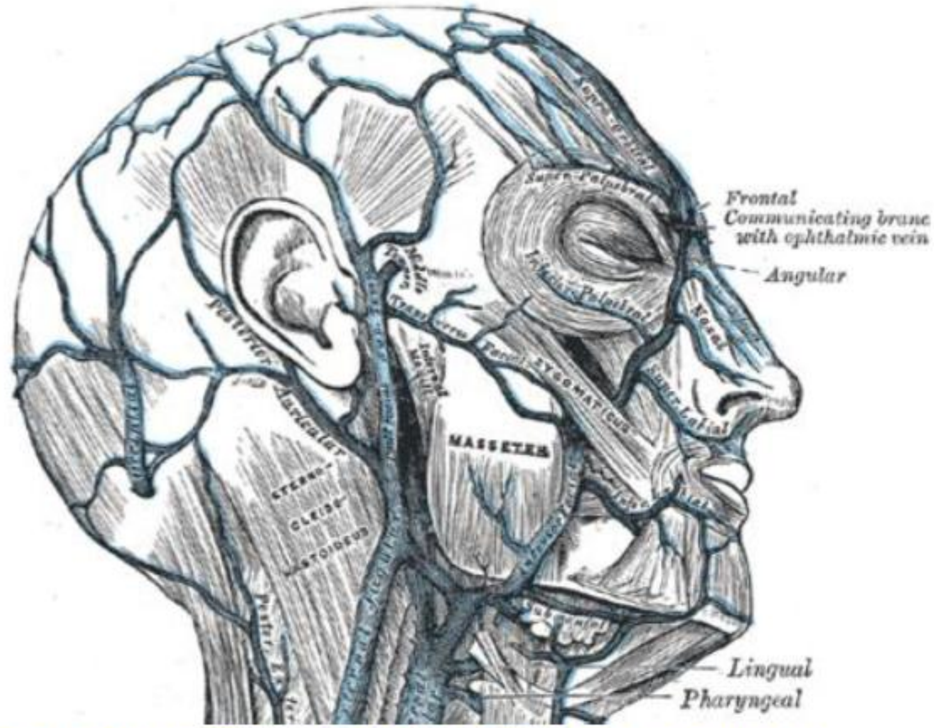
- *facial*
  - *angular*
  - *lateral nasal*
  - *superior labial*
- *infraorbital*

ICA

- *Branches of the ophthalmic a.*

VENOUS DRAINAGE

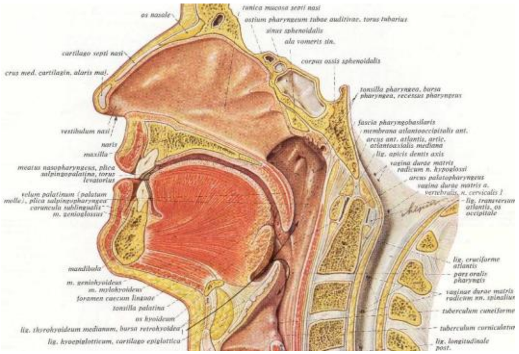




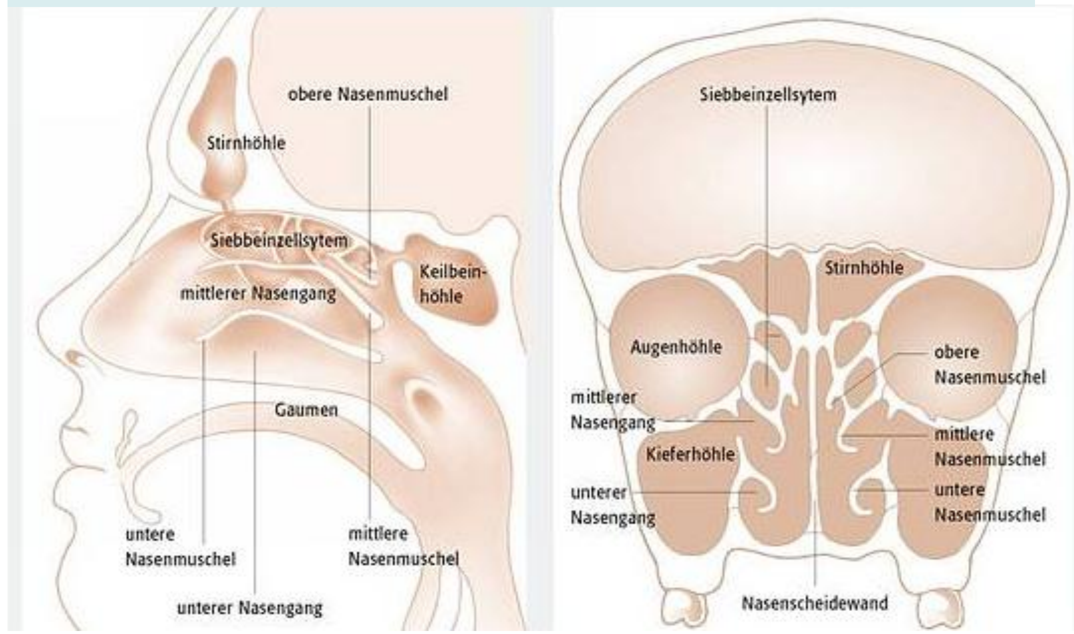
Veins in the nose essentially follow the arterial pattern. They are significant for their direct communication with the cavernous sinus and for their lack of valves; these features potentiated the intracranial spread of infection. Even with the abundant blood supply of the nose, smoking does compromise postoperative healing**.**

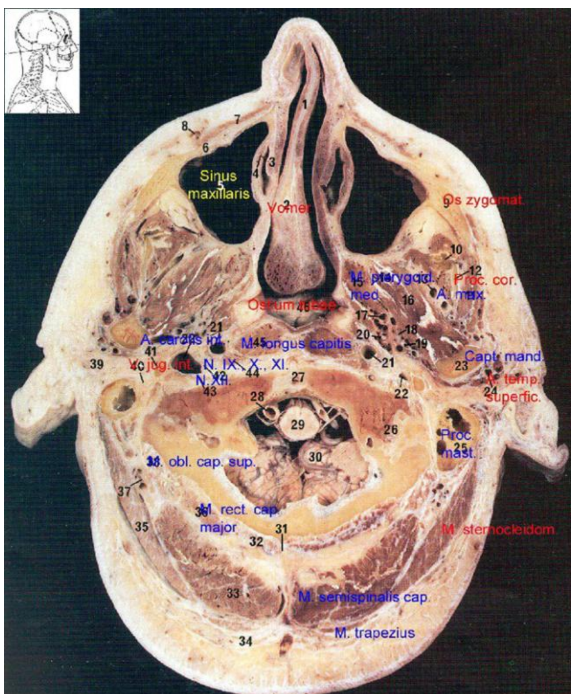
IMPORTANT ANASTOMOSIS:
Between the ***angular vein*** and the ***inferior ophthalmic vein*** - a direct conduit towards the **cavernous sinus**

TOPOGRAPHY OF THE NASAL CAVITY



LIMITED BY 4 WALLS: SUP, INF, LAT, MED



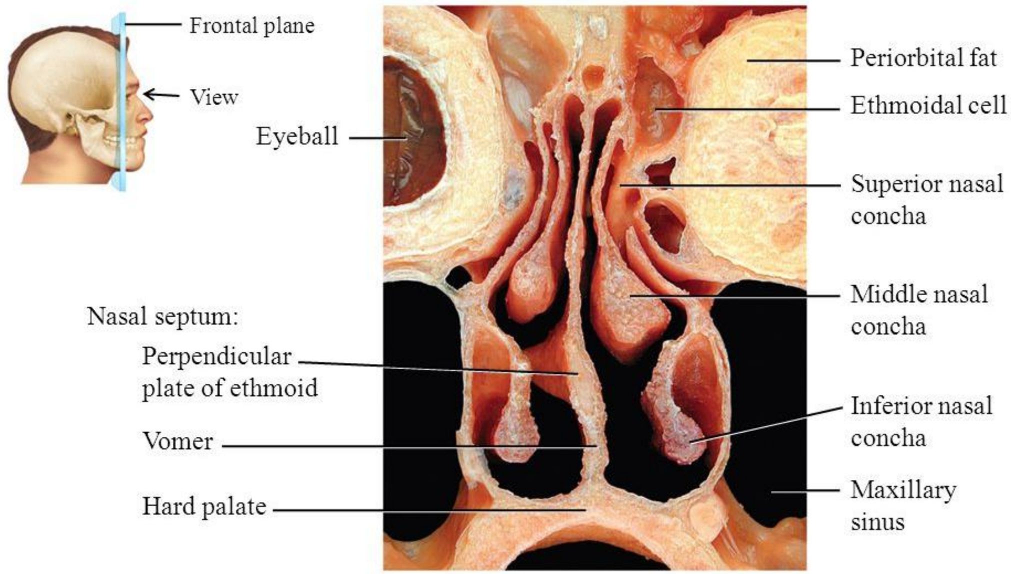


Vicinity of other cavities of the viscerocranium or neurocranium

Shared borders

SUP	**cribriform lamina**
LAT	**ethmoid air cells / orbital plate**
INF	**hard palate**
MED	**nasal septum between the two halves**

DIVISIONS OF THE NASAL CAVITY



(c) Frontal section showing conchae

Dissection Shawn Miller, Photograph Mark Nielsen

(Nostrils/piriform aperture)

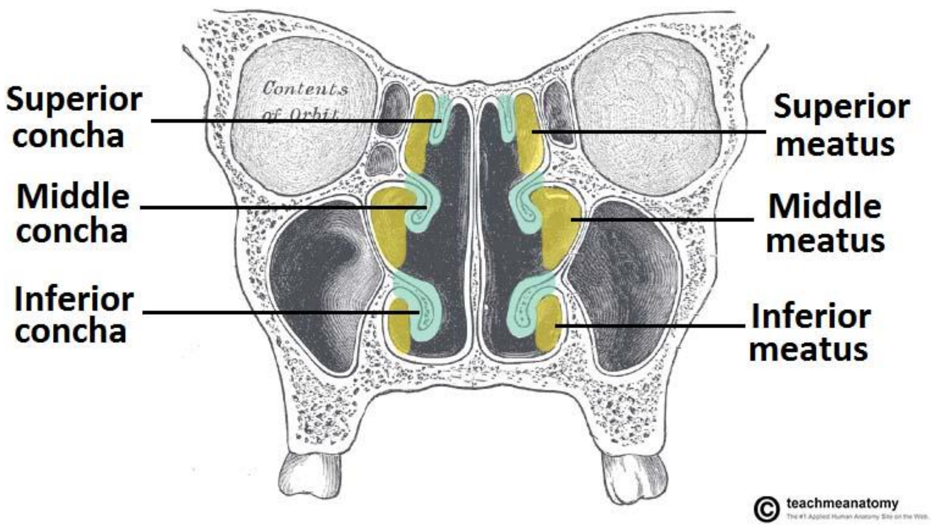
Nasal vestibule (*vibrissae*)

limen nasi

Nasal cavity proper

- olfactory region
- respiratory region

(*Choanae*)



MED

Common nasal meatus

(incl the spenoethmoidal recess)

LAT

Superior nasal meatus

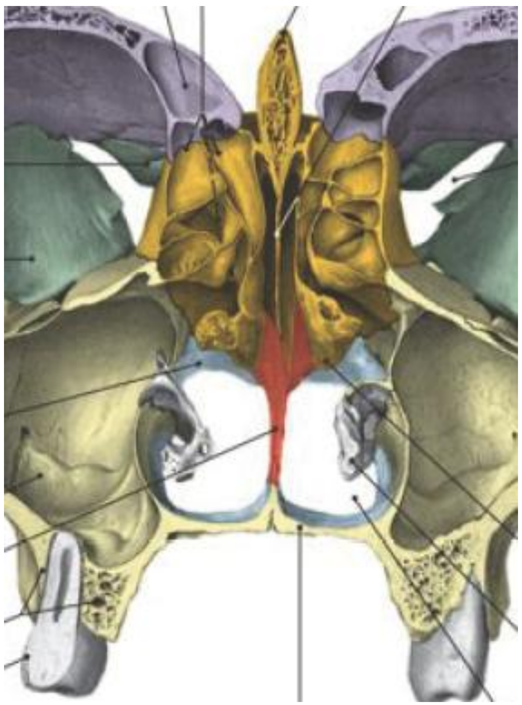
Middle nasal meatus

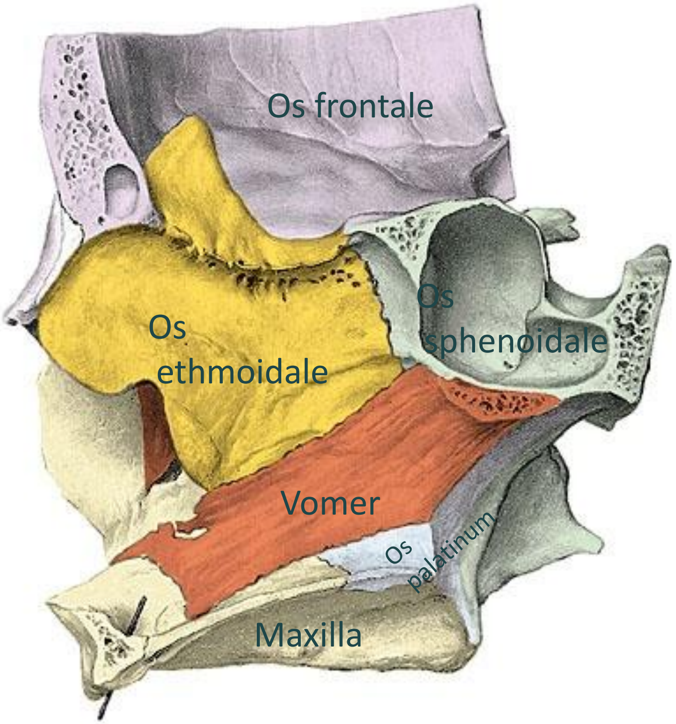
Inferior nasal meatus

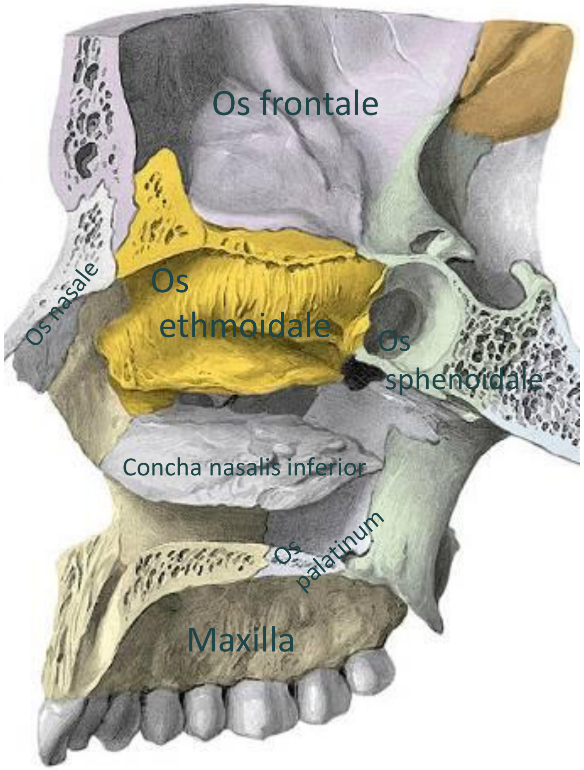
NASAL CAVITY – INTERNAL WALL STRUCTURE

Bony nasal cavity

(Revision from the 1st semester)

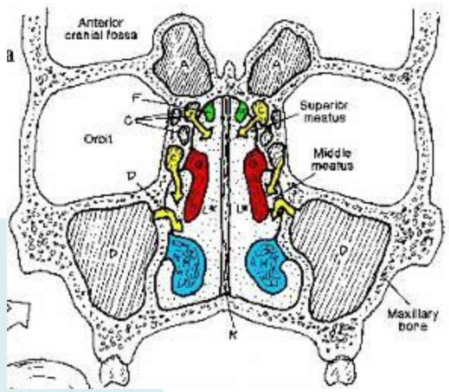


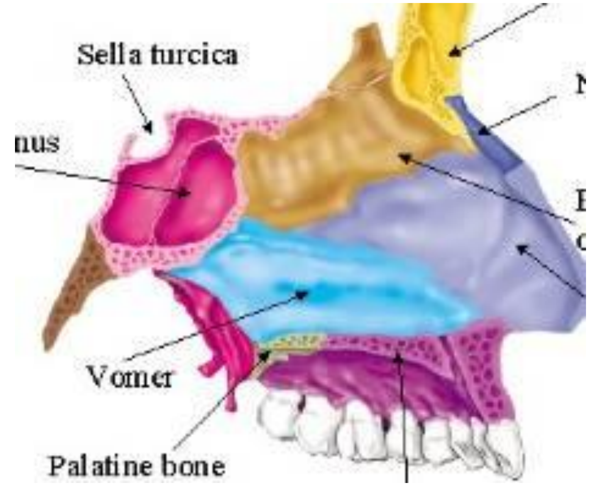




THE NASAL CAVITY or NASAL CAVITIES

Separated by the nasal
septum - 2 halves





MEDIAL,
UNDIVIDED PART
COMMON NASAL MEATUS

LIMITED BY Hard palate
Cribriform lamina
Nasal septum

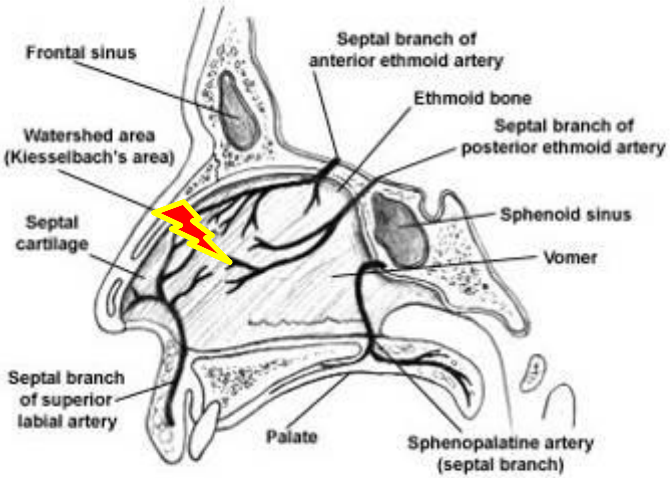
CONNECTIONS
- Piriform aperture
- cribriform lamina/foramina
- Incisive canal
- Sphenopalatine foramen
- Aperture of the sphenoidal sinus
- choanae

NASAL SEPTUM
(bone, cartilage, CT)

Composed of the
vomer
ethmoid /perpendicular plate
septal cartilage
Mucosal covering: respiratory epith. upon lamina propria (rich vascular supply)

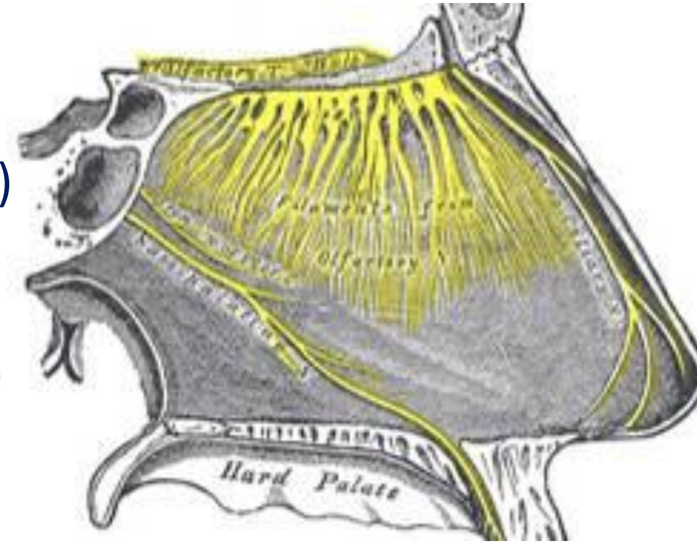
pars mobilis *(ant portion)*
Covered by skin

NASAL SEPTUM (MEDIAL WALL)



INNERVATION

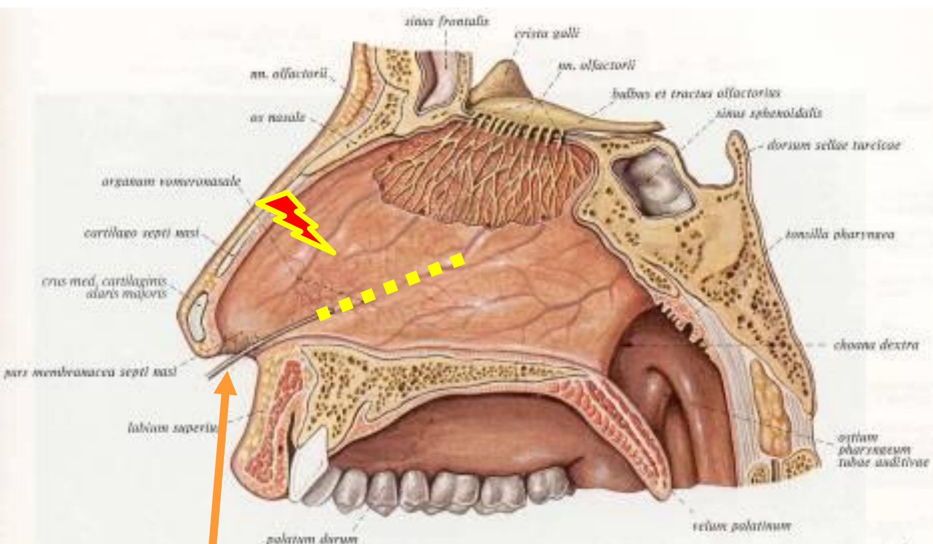
Medial (or septal) branches of the **nasociliary** and the **nasopalatine** nerves



BLOOD SUPPLY

- sphenopalatine a
- anterior and posterior ethmoid aa
- superior labial a (anteriorly)
- greater palatine a (posteriorly).

The Kiesselbach plexus, or the Little area, *represents a region in the anteroinferior third of the nasal septum, where all 3 of the chief blood supplies to the internal nose converge.*

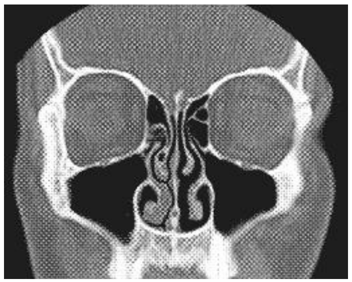


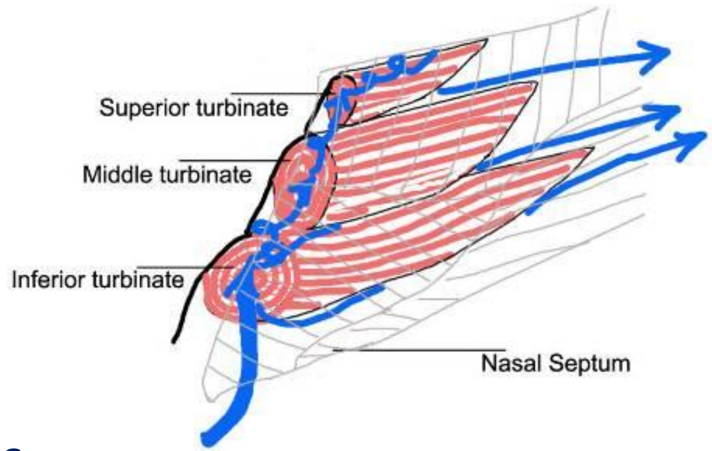
Vomeronasal organ of Jacobson

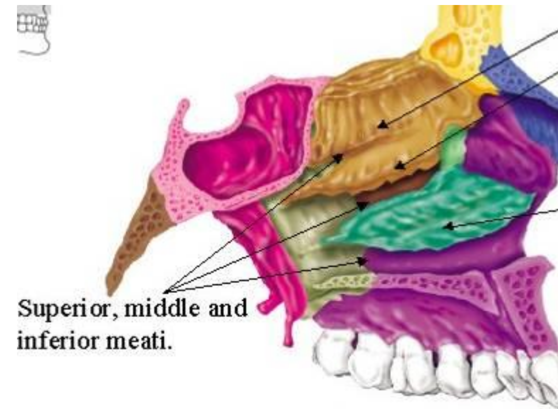
NASAL CAVITY (LATERAL WALL)

SUPERIOR, MIDDLE AND INFERIOR MEATUS 3 conchae – 3 corresponding meatus

- leading the air IN and airing/draining the paranasal sinuses; or draining the orbit







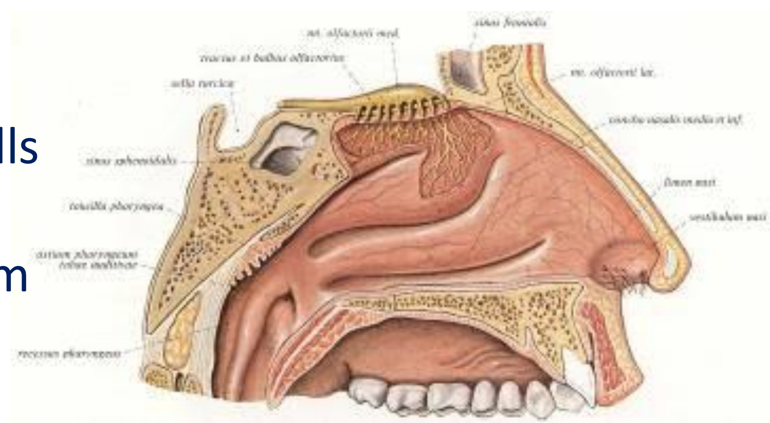
CONNECTIONS

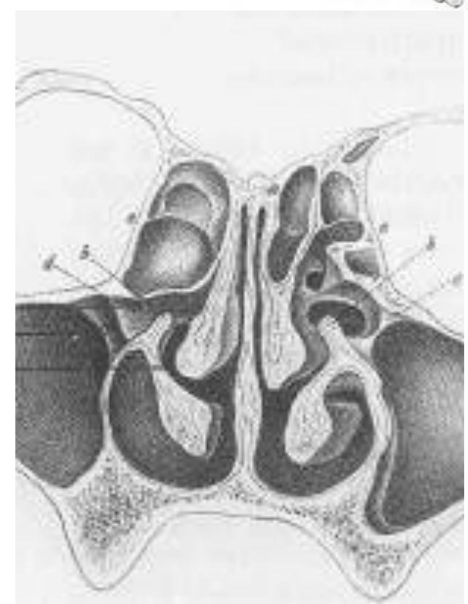
Superior nasal meatus

- Post ethm air cells

Middle nasal meatus

- ant, med ethm air cells
- semilunar hiatus
- Ethmoid infundibulum
- Semilunar hiatus





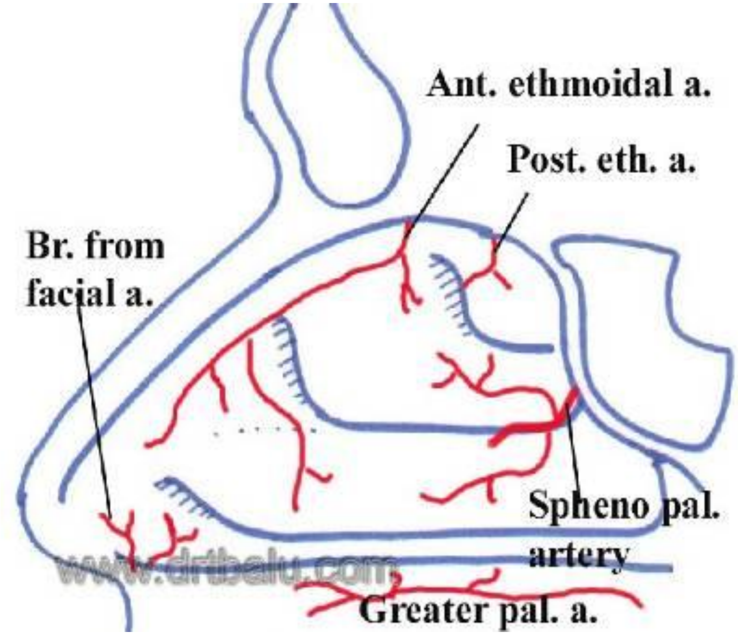
Inferior nasal meatus

- nasolacrimal canal (Hasner valve)

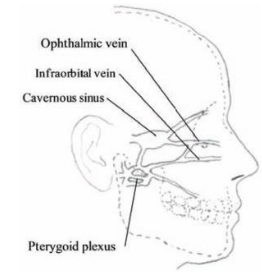
NASAL CAVITY (LATERAL WALL)

BLOOD SUPPLY

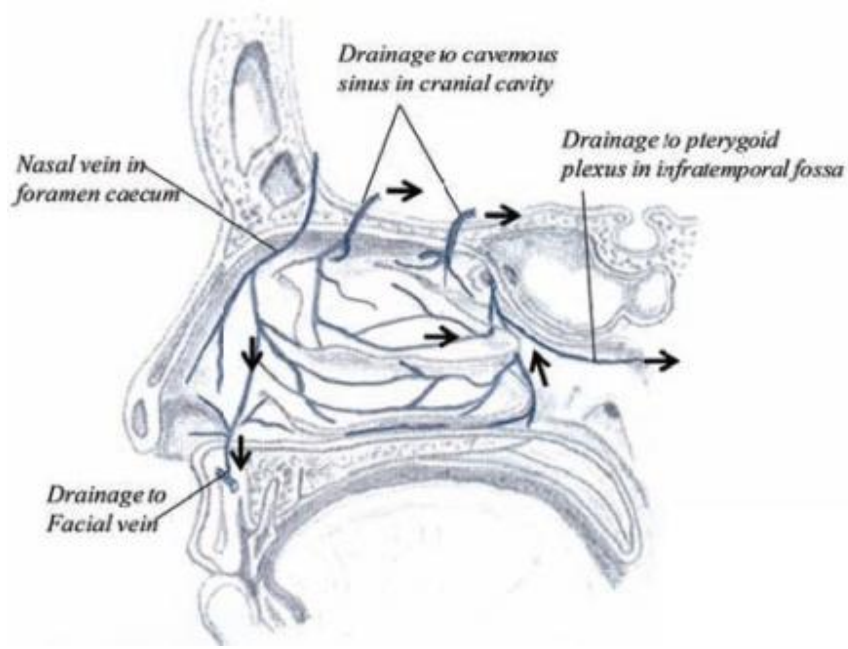
Arteries



Veins



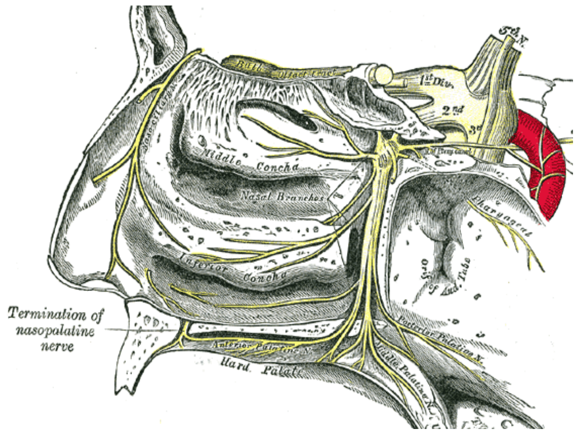
pterygoid and pharyngeal plexus
plexus cavernosi concharum

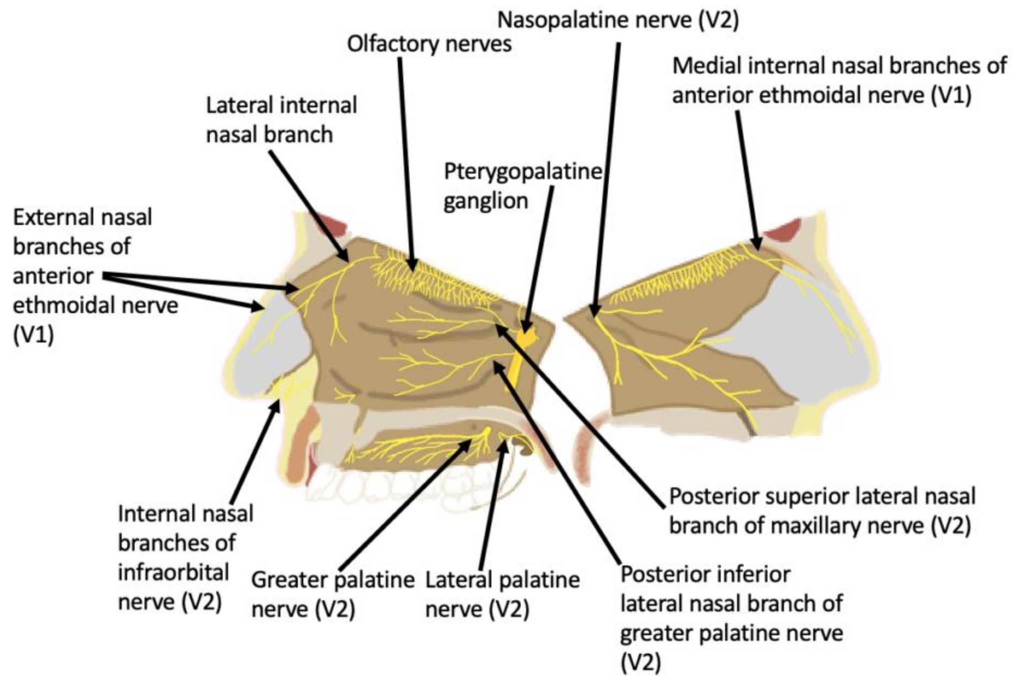


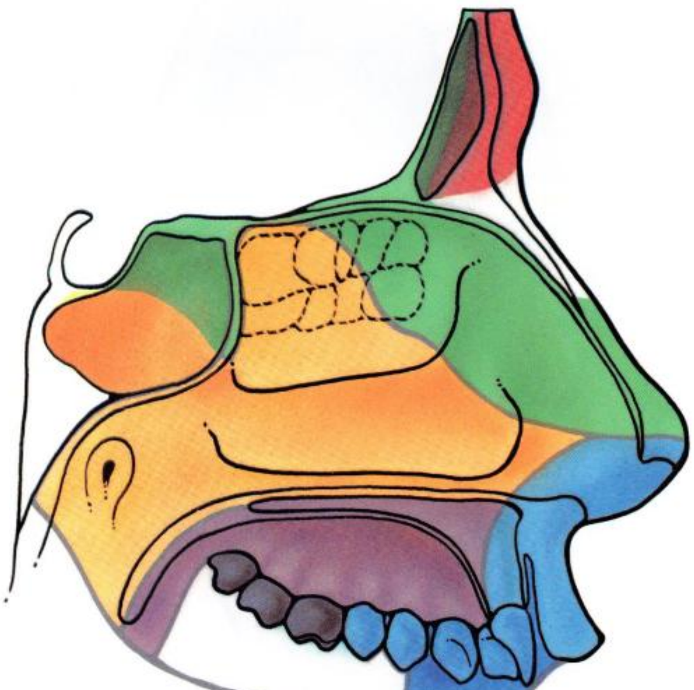
Posteroinferior: **sphenopalatine a**
superior: **ant and post ethmoid aa.**
Inferior: **facial a.**

NASAL CAVITY (LATERAL WALL)

INNERVATION







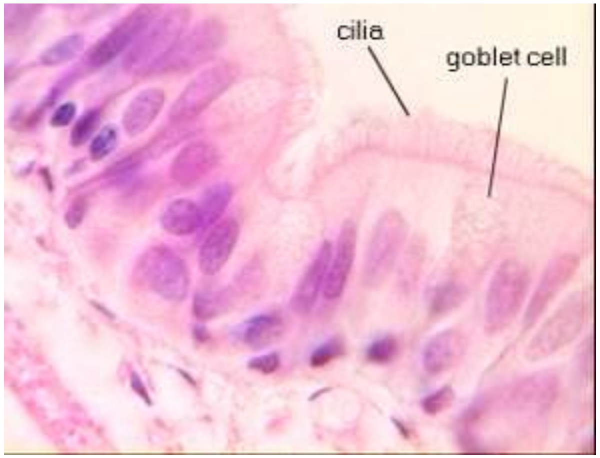
Surgical anatomy of the face

Ant ethmoidal nerve
Sphenopalatine nerve
Infraorbital nerve
Ant/med sup alveolar nerve
Post sup alveolar nerve

HISTOLOGICAL FEATURES

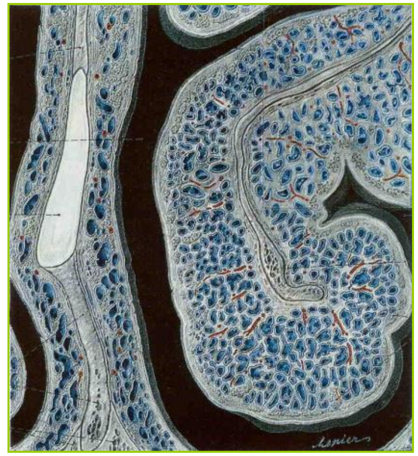
Just for information

Nasal mucosa



- *Thickness: 8 –12 µm*
- *Transport speed: 3 – 12 mm/s*
- *Daily secretory production: 200 g*
- *Emptying of all paranasal sinuses towards the pharynx*
- *Funktion of the nasal mucosa:*
  - *defence machanisms (immune system)*
  - *moistenig and cleaning the air*
  - *olfaction*

Nasal concha



Semicavernous structure

(***Stuffy nose***)

Olfactory epithelium

Neuroepithelial cells . receptors